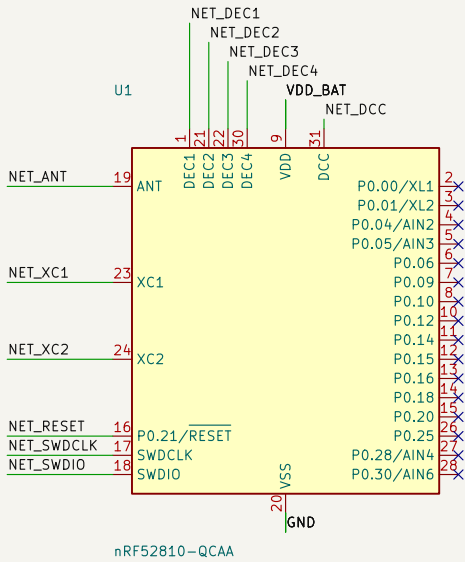
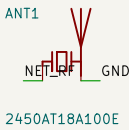
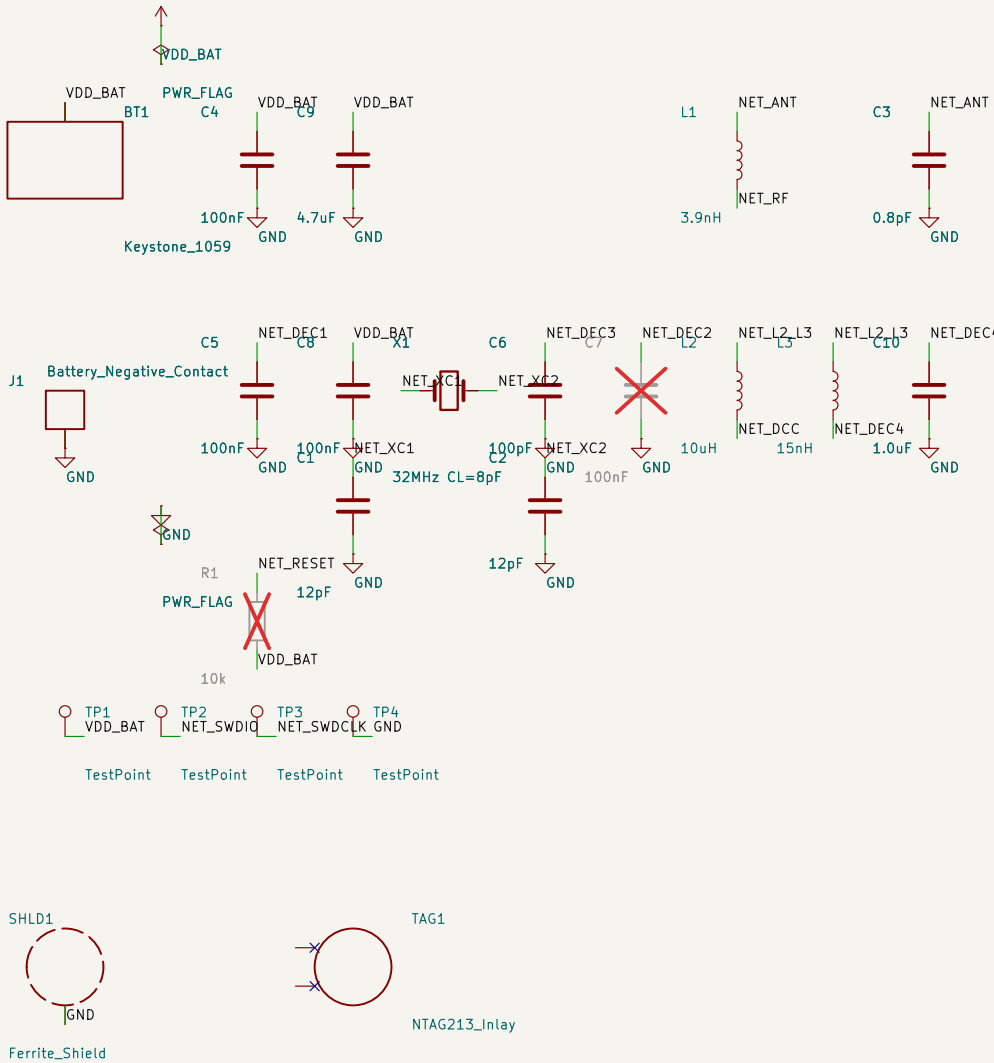




NOTA 1: Pinagem U1 (QFN32 5x5mm) verificada pino a pino no datasheet oficial Nordic

nRF52810 PS, Table 99 (QFN32 pin assignments): DEC1=1, VDD=9/25/32 (tres pinos).

P0.21/nRESET=16, SWDCLK=17, SWDIO=18, ANT=19, VSS=20/29+die pad, DEC2=21, DEC3=22,

XC1=23, XC2=24, DEC4=30, DCC=31. Fonte: docs.nordicsemi.com/bundle/ps_nrf52810

(PDF nRF52810_PS, secao 7.1.2). NAO usar mapeamento QFN48 (pinos 33-43/VDDH).

NOTA 2: Casamento RF: ANT(19) -> no com C3 0.8pF shunt->GND -> L1 3.9nH serie -> ANT1 (2 elementos).

Ajuste fino obrigatorio com VNA na placa real (Nordic/Johanson: valores de EVB).

Topologia conforme nRF52810 PS Fig. 146/Tabela 123 (corrigida vs netlist_conexoes.md em 2026-09-09; ver docs/verificacao_matematica.md claims 18/20).

NOTA 3: Desacoplos/ladder DC/DC conforme PS Fig. 146/Tabela 123: DEC1(1): C5 100nF;

DEC3(22): C6 100pF; DEC2(21): C7 100nF DNP (pos. N.C. da ref.); VDD(9/25/32): C4+C8 100nF + C9 4.7uF; DCC(31) -L2 10uH- L3 15nH- DEC4(30) + C10 1.0uF.

NOTA 4: 16 GPIOs nao usados pelo firmware: NC explicito (pinos 2-8, 10-15, 26-28).

NOTA 5: Layout: keep-out de cobre sob ANT1 e rede de casamento RF; plano GND contiguo na camada 2:

die pad (33/VSS) com multiplas vias para GND; ferrite SHLD1 entre BT1 e TAG1;

PCB 2 camadas 0.8mm. Cristal 32MHz CL=8pF com C1/C2 12pF NP0 (Ccap=2*CL-4pF).

NOTA 6: R1 10k pull-up nRESET esta DNP: o nRF52810 tem pull-up interno (~13k); popular

apenas se houver ruido forte na linha de reset.

NOTA 7: ERC (projeto): pin_not_driven=ignore (nets XC1/XC2/SWDCLK/DEC* sao passivas por

design, sem driver a bordo - ref Nordic); footprint_link_issues=ignore (kicad-cli

10.0.6 nao carrega fp-lib-table; links de footprint validados por nome).

BOM: BOM_discreta_corrigida_nRF52810_QFN32.csv		
Pinagem QFN32: Nordic nRF52810 PS Table 99 - VDD=9/25/32 DEC4=30 DCC=31		
Variante B - rota discreta nRF52810-QCAA QFN32 5x5mm (sem modulo certificado)		
NibasRockBar		
Sheet: /		
File: wristband_discreto.kicad_sch		
Title: Wristband Discreta nRF52810		
Size: A3	Date: 2026-09-09	Rev: A
KiCad E.D.A. 10.0.6		Id: 1/1